

PROJECT TOPIC: NATIONAL PARKS OF MAHARASHTRA

Topic Description: Comprehensive Tabular Study of Five National Parks of Maharashtra: Geography, Forest Biomes, Flagship Wildlife, and Conservation Challenges

In the framework of national wildlife conservation, National Parks represent the highest, most stringent category of statutory protected areas notified under Section 35 of the Wildlife (Protection) Act, 1972. Unlike Wildlife Sanctuaries, where limited human activities such as regulated livestock grazing or minor forest produce collection may be permitted by the Chief Wildlife Warden, a National Park is an inviolate core sanctuary where zero human exploitation, grazing, forestry operations, or private land ownership is legally permitted.

This project presents a comprehensive ecological and comparative investigation of **Five National Parks of Maharashtra: Tadoba-Andhari National Park (Chandrapur), Sanjay Gandhi National Park (Mumbai/Thane), Gugamal National Park (Amravati - Melghat), Navegaon National Park (Gondia), and Chandoli National Park (Sangli/Satara/Kolhapur/Ratnagiri).**

Maharashtra possesses an extraordinary ecological diversity, spanning the tropical moist evergreen forests of the Western Ghats to the dry teak and bamboo forests of the Vidarbha basin. By analyzing their geographic locations, administrative areas, floral biomes, flagship mammalian fauna, and pressing anthropogenic threats, this study evaluates the crucial role these parks play in safeguarding Maharashtra's natural heritage.

2. Importance of the topic (विषयाचे महत्त्व)

National Parks constitute the irreplaceable biological backbone of biodiversity conservation in Maharashtra:

Key Dimensions of National Park Importance :-

- **Apex Predator & Umbrella Species Protection:** Parks like Tadoba and Gugamal protect critical breeding populations of Royal Bengal Tigers and Indian Leopards. By safeguarding these apex predators, the entire underlying trophic pyramid of herbivores, flora, and soil microbes is automatically protected.
- **Catchment Security for Major Rivers:** National Parks safeguard the origin basins and catchments of Maharashtra's lifeline rivers. Chandoli secures the catchment of the Varna river, while Sanjay Gandhi National Park protects the catchments of Vihar and Tulsi lakes, supplying clean drinking water to Mumbai.
- **Gene Banks for Climate Adaptation:** Undisturbed wildernesses act as evolutionary gene banks preserving wild relatives of cultivated crops, indigenous medicinal lianas, and endangered timber trees.
- **Carbon Sequestration & Microclimate Regulation:** Dense, mature climax canopies sequester millions of tons of carbon dioxide, functioning as massive regional carbon sinks and cooling urban centers.

National Parks ensure that evolutionary natural processes continue uninterrupted by human commercial greed.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The comparative investigation was executed with structured ecological, geographical, and conservation objectives:

Structured Project Objectives :-

- **Statutory & Geographic Profiling:** To compile an official gazette profile of five National Parks in Maharashtra, documenting district locations, total protected area (sq km), and year of notification.
- **Forest Biome & Floral Characterization:** To classify the distinct vegetation types (Tropical Dry Deciduous, Moist Deciduous, Semi-Evergreen) and dominant tree species across each park.
- **Flagship & Endemic Faunal Inventory:** To identify the primary mammalian, avian, and reptilian wildlife populations sheltered within each protected core.
- **Threat & Conflict Assessment:** To evaluate park-specific conservation challenges, including human-wildlife conflicts, highway/railway corridor fragmentation, poaching, and eco-tourism pressures.
- **Student Eco-Tourism Principles:** To formulate ethical guidelines for responsible student wildlife visits and wildlife habitat stewardship.

The scope encompasses forest department census reports, gazette notifications, and biogeographic atlas data.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A structured secondary research, cartographic review, and analytical survey methodology was deployed:

Methodological Research Protocol :-

- **Phase 1 – Official Gazette & Census Data Review:** Consulted administrative wildlife status reports published by the Maharashtra Forest Department (PCCF Wildlife, Nagpur) and the National Tiger Conservation Authority (NTCA).
- **Phase 2 – Geographic & Biome Mapping:** Mapped the geographic distribution of all five parks across Maharashtra's distinct agro-climatic zones (Western Ghats, Deccan Plateau, Vidarbha).
- **Phase 3 – Tabular Parameter Standardization:** Standardized empirical parameters for comparison: Park Name, District, Area (sq km), Forest Type, Dominant Flora, Flagship Fauna, and Special Features.
- **Phase 4 – Threat Analysis:** Evaluated documented conflict reports, specifically urban leopard interactions around Sanjay Gandhi National Park and tiger corridor bottlenecks around Tadoba.
- **Phase 5 – Synthesis & Syllabus Alignment:** Correlated findings with Chapter 7 competencies under the Maharashtra State Board Standard XI EVS journal.

All recorded statistical metrics were authenticated against official government wildlife gazetteers.

5. Observations (निरीक्षणे)

Systematic compilation of official wildlife data yielded a comprehensive comparative matrix of Maharashtra's five National Parks:

5.1 Tabular Record of Five National Parks of Maharashtra

National Park Name	District & Area	Forest Biome Type	Common Flora & Fauna	Special Ecological Feature
Tadoba-Andhari (Est. 1955)	Chandrapur (625.4 sq km)	Southern Tropical Dry Deciduous forest	Flora: Teak, Ain, Bija, Bamboo. Fauna: Royal Bengal Tiger, Leopard, Dhole, Gaur, Marsh Crocodile.	Known as the 'Jewel of Vidarbha'; highest tiger density in Maharashtra; prime breeding source population for central India.
Sanjay Gandhi (SGNP) (Est. 1983)	Mumbai Suburban & Thane (103.8 sq km)	Tropical Moist and Dry Deciduous forest	Flora: Kadamba, Teak, Karanj, Shisham. Fauna: Indian Leopard, Spotted Deer, Rusty-spotted Cat, Atlas Moth.	Unique wilderness surrounded by a mega-metropolis; shelters Kanheri Caves; highest urban leopard density in the world.
Gugamal (Melghat) (Est. 1975)	Amravati (Satpura range) (361.3 sq km)	Southern Dry Deciduous; rugged hilly terrain	Flora: Teak, Mahua, Dhawada, Haldu. Fauna: Tiger, Sloth Bear, Chausingha (Four-horned Antelope), Caracal.	Core zone of Melghat Tiger Reserve; critical hilly watershed for the Tapi river basin; home to the rare Forest Owlet.
Navegaon	Gondia	Mixed Dry	Flora: Haldu, Jamun, Mahua, Kawat. Fauna: Leopard,	Centrally located historic Navegaon lake built in the 18th

6. Analysis of data (माहितीचे विश्लेषण)

Detailed comparative analysis of Maharashtra's National Parks illustrates vital ecological dynamics and severe landscape threats:

Core Analytical Inferences :-

- **Urban-Wild Interface Pressures (The SGNP Case):** Sanjay Gandhi National Park is an extraordinary global anomaly: a 104 sq km wild forest completely surrounded by 20 million urban residents. The presence of over 45 resident leopards feeding on stray dogs and feral pigs without attacking humans daily proves remarkable behavioral adaptation, but requires strict garbage management to avoid fatal encounters.
- **Corridor Fragmentation as the Primary Extinction Threat:** Tigers in Tadoba cannot survive as an isolated genetic island. Linear infrastructure projects—such as widening National Highway 44 and rail line doublings—sever historic dispersal corridors between Tadoba, Nagzira, and Kanha, leading to fatal road accidents and inbreeding depression.
- **Watershed and Climatological Valuation:** Chandoli and Gugamal protect dense mountain catchments that feed agricultural reservoirs downstream. Deforesting these catchments would trigger severe siltation of dams and disastrous summer water shortages across western Maharashtra and Vidarbha.
- **Eco-Tourism Bottlenecks:** Unregulated safari vehicle traffic in Tadoba causes dust pollution, disrupts nocturnal predator hunting cycles, and creates human disturbances along core waterholes during peak summer dry months.

The analysis affirms that wildlife conservation requires landscape-scale connectivity, not just isolated fortress parks.

7. Results and Conclusions (निष्कर्ष)

Conclusion: Maharashtra's five National Parks are vital ecological sanctuaries that preserve the state's biological heritage, regulate river catchments, and shelter endangered wildlife. Ensuring their long-term survival demands wildlife corridor protection, mitigated linear infrastructure, and active community-inclusive buffer zone management.

Action Checklist for Responsible Wildlife Citizenship :-

- ❑ Advocate for scientific wildlife overpasses and underpasses along all highways crossing forest corridors.
- ❑ Follow strict eco-tourism ethics: maintain absolute silence, never litter plastic, and stay inside safari vehicles.
- ❑ Support community forest rights and eco-development committees (EDCs) in fringe villages around parks.
- ❑ Never purchase wild animal products, skins, bones, feathers, or illegal wildlife souvenirs.
- ❑ Spread awareness among urban peers regarding non-violent coexistence with urban leopards around SGNP.

8. Bibliography / References (संदर्भ सूची)

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